

Synthetic network

Six policies, two planning modes. 150 simulated vehicles; seed 7. Time is in abstract ticks.

Policy	Planning	Mean (ticks)	P95 (ticks)	Delay vehicle-ticks	Peak load	Replans	Changes
Static	Departure	53.67	105.00	7224.0	35.33x	0	0
Reactive	Departure	18.30	40.00	1675.0	14.00x	0	0
Predictive	Departure	18.30	40.00	1675.0	14.00x	0	0
Anticipatory	Departure	10.68	14.00	473.0	4.00x	0	0
Cooperative	Departure	9.89	14.00	298.0	3.33x	0	0
RL swarm	Departure	9.98	13.00	330.0	3.33x	0	0
Static	Junction	53.67	105.00	7224.0	35.33x	120	0
Reactive	Junction	14.71	25.00	1084.0	6.00x	39	24
Predictive	Junction	13.55	24.00	902.0	5.50x	80	54
Anticipatory	Junction	10.68	14.00	473.0	4.00x	31	4
Cooperative	Junction	9.89	14.00	298.0	3.33x	58	0
RL swarm	Junction	10.00	14.00	326.0	3.00x	26	11

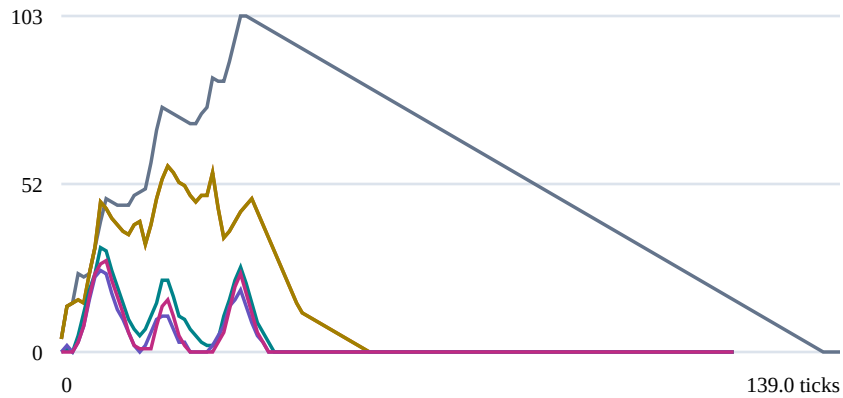
Replans count junction decisions with multiple legal candidates; changes count decisions that revise the remaining route. All trip metrics include background vehicles. Peak load is occupancy divided by capacity times free-flow duration.

Limitations: point queues with unlimited storage; no lane or signal model and no live traffic feed. RL weights were trained for departure decisions on the corridor network. Junction and synthetic-network results reuse these weights without retraining; these runs are not a new held-out evaluation.

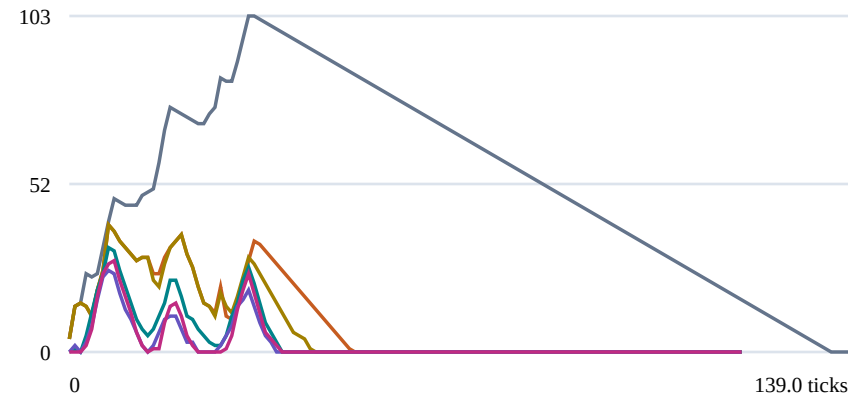
Synthetic network / Traffic over time

Static Reactive Predictive Anticipatory Cooperative RL swarm

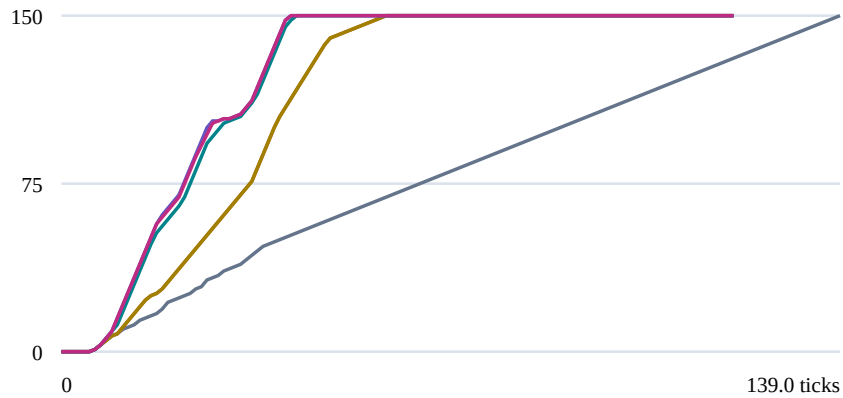
Vehicles waiting / Departure



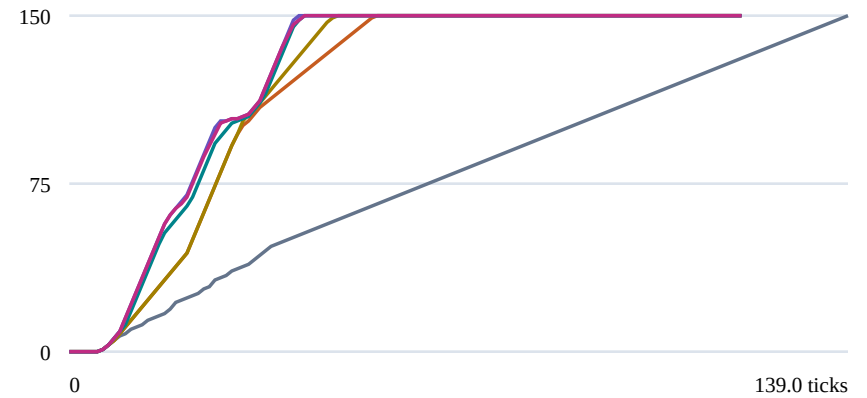
Vehicles waiting / Junction



Completed trips / Departure



Completed trips / Junction



Both columns use identical axes. Curves may overlap where policies produce the same outcome. Values come directly from the recorded simulator runs; the PDF does not rerun or change the experiment.

Milpitas - Sunnyvale: Light commute

Six policies, two planning modes. 72 simulated vehicles; seed 19. Real-road geometry; simulated traffic. One simulator tick is 10 seconds.

Policy	Planning	Mean (min)	P95 (min)	Delay vehicle-min	Peak load	Replans	Changes
Static	Departure	15.85	16.00	1.5	1.00x	0	0
Reactive	Departure	15.85	16.00	1.5	1.00x	0	0
Predictive	Departure	15.85	16.00	1.5	1.00x	0	0
Anticipatory	Departure	15.85	16.00	1.5	1.00x	0	0
Cooperative	Departure	15.85	16.00	1.5	1.00x	0	0
RL swarm	Departure	16.10	16.00	1.5	1.00x	0	0
Static	Junction	15.85	16.00	1.5	1.00x	120	0
Reactive	Junction	15.85	16.00	1.5	1.00x	120	0
Predictive	Junction	15.85	16.00	1.5	1.00x	120	0
Anticipatory	Junction	15.85	16.00	1.5	1.00x	120	0
Cooperative	Junction	15.85	16.00	1.5	1.00x	120	0
RL swarm	Junction	17.73	21.83	1.5	1.00x	103	33

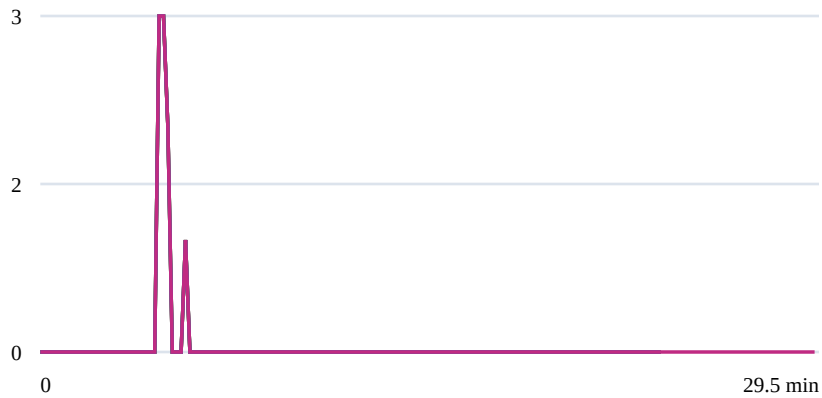
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Limitations: point queues with unlimited storage; no lane or signal model and no live traffic feed. RL weights were trained for departure decisions on the corridor network. Junction and synthetic-network results reuse these weights without retraining; these runs are not a new held-out evaluation.

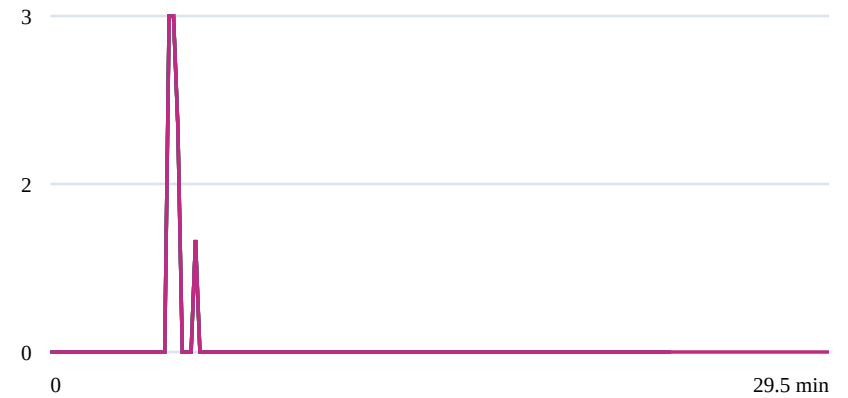
Milpitas - Sunnyvale: Light commute / Traffic over time

Static Reactive Predictive Anticipatory Cooperative RL swarm

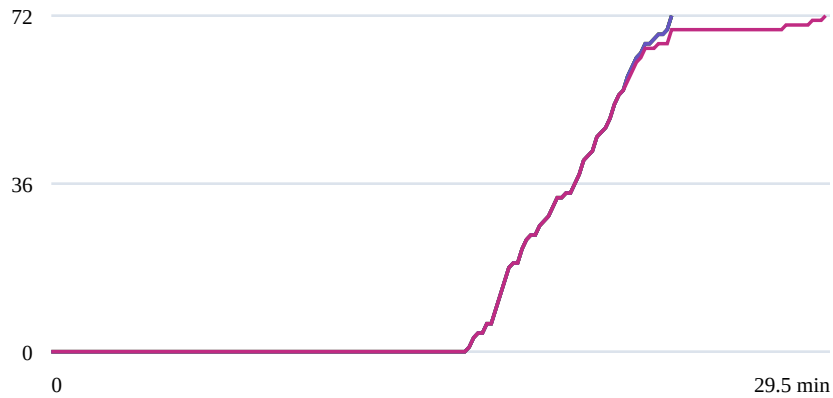
Vehicles waiting / Departure



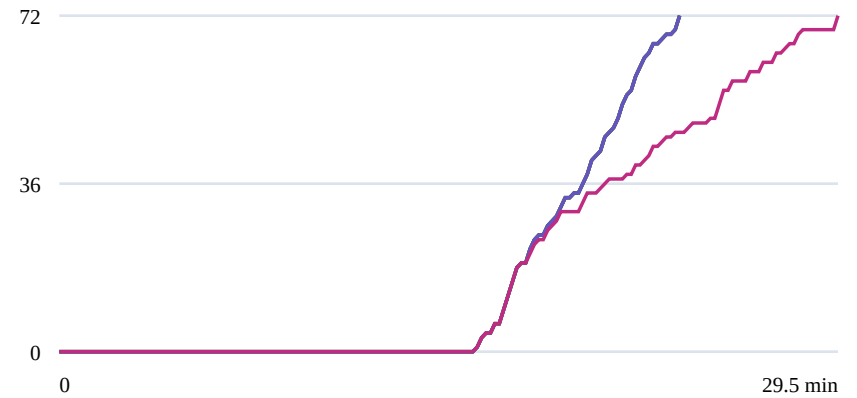
Vehicles waiting / Junction



Completed trips / Departure



Completed trips / Junction



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Milpitas - Sunnyvale: Departure surge

Six policies, two planning modes. 210 simulated vehicles; seed 19. Real-road geometry; simulated traffic. One simulator tick is 10 seconds.

Policy	Planning	Mean (min)	P95 (min)	Delay vehicle-min	Peak load	Replans	Changes
Static	Departure	20.13	24.17	902.2	11.40x	0	0
Reactive	Departure	20.13	24.17	902.2	11.40x	0	0
Predictive	Departure	20.13	24.17	902.2	11.40x	0	0
Anticipatory	Departure	20.13	24.17	902.2	11.40x	0	0
Cooperative	Departure	20.13	24.17	902.2	11.40x	0	0
RL swarm	Departure	20.13	24.17	902.2	11.40x	0	0
Static	Junction	20.13	24.17	902.2	11.40x	350	0
Reactive	Junction	20.13	24.17	902.2	11.40x	350	0
Predictive	Junction	20.13	24.17	902.2	11.40x	350	0
Anticipatory	Junction	20.13	24.17	902.2	11.40x	350	0
Cooperative	Junction	20.13	24.17	902.2	11.40x	350	0
RL swarm	Junction	22.28	30.00	902.2	11.40x	282	156

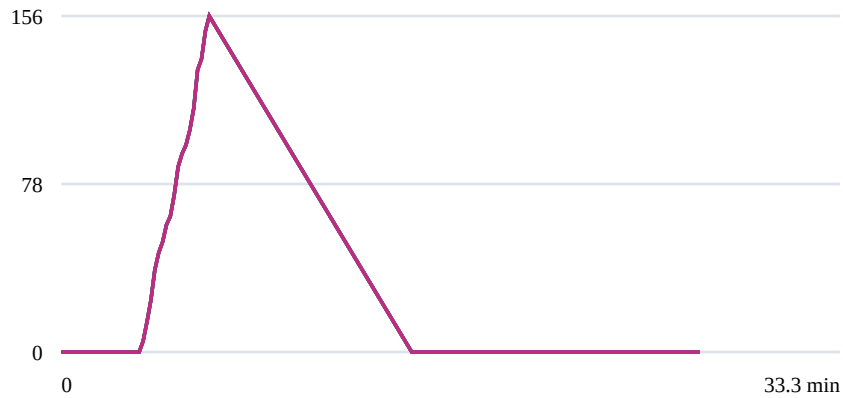
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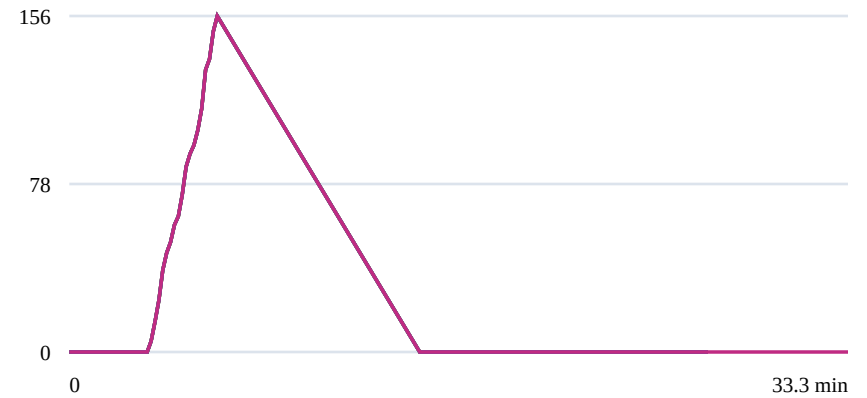
Milpitas - Sunnyvale: Departure surge / Traffic over time

Static Reactive Predictive Anticipatory Cooperative RL swarm

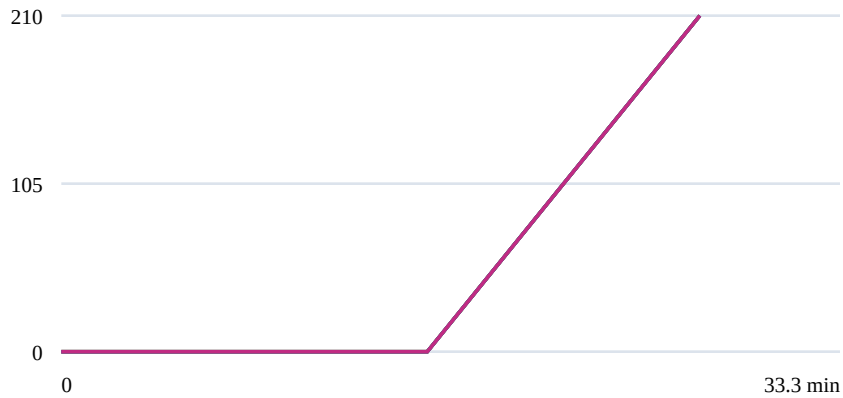
Vehicles waiting / Departure



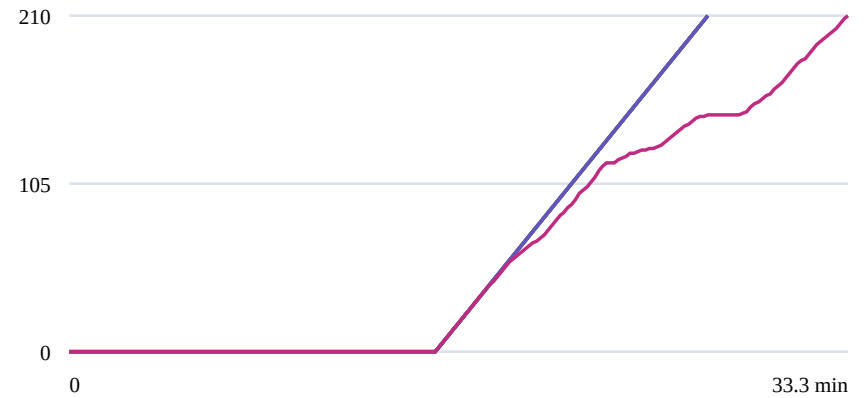
Vehicles waiting / Junction



Completed trips / Departure



Completed trips / Junction



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Milpitas - Sunnyvale: CA 237 capacity stress

Six policies, two planning modes. 210 simulated vehicles; seed 19. Real-road geometry; simulated traffic. One simulator tick is 10 seconds.

Policy	Planning	Mean (min)	P95 (min)	Delay vehicle-min	Peak load	Replans	Changes
Static	Departure	31.80	46.33	3352.2	24.33x	0	0
Reactive	Departure	31.80	46.33	3352.2	24.33x	0	0
Predictive	Departure	31.80	46.33	3352.2	24.33x	0	0
Anticipatory	Departure	23.48	27.67	1169.0	11.40x	0	0
Cooperative	Departure	22.87	27.67	1042.0	11.40x	0	0
RL swarm	Departure	22.65	27.50	926.0	11.40x	0	0
Static	Junction	31.80	46.33	3352.2	24.33x	350	0
Reactive	Junction	23.76	28.17	1253.0	11.40x	350	162
Predictive	Junction	24.24	29.67	1395.2	11.40x	350	101
Anticipatory	Junction	23.48	27.67	1169.3	11.40x	350	34
Cooperative	Junction	22.45	26.67	935.2	11.40x	350	45
RL swarm	Junction	23.55	30.17	955.3	11.40x	293	123

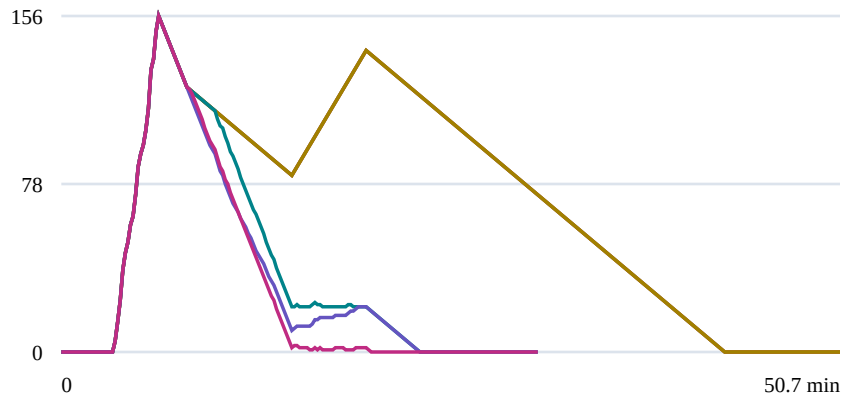
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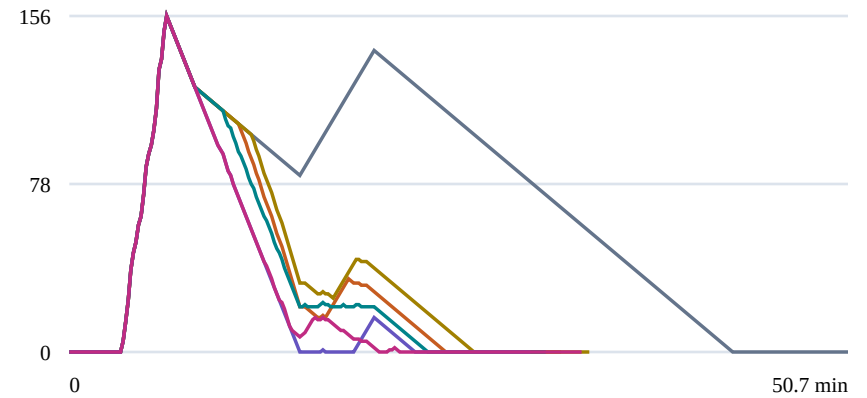
Milpitas - Sunnyvale: CA 237 capacity stress / Traffic over time

Static Reactive Predictive Anticipatory Cooperative RL swarm

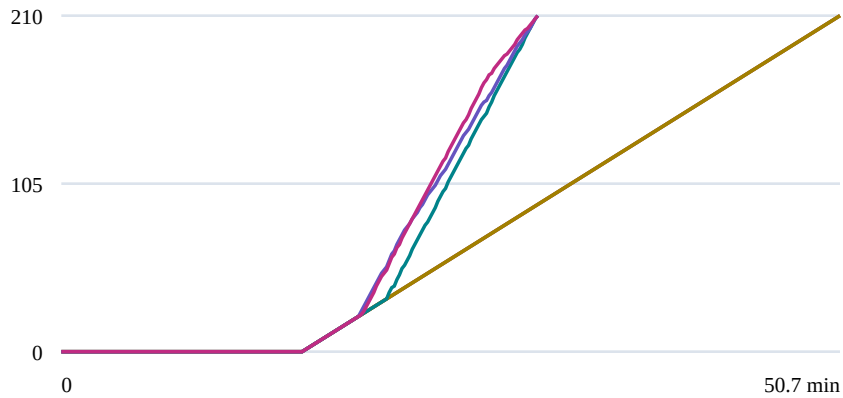
Vehicles waiting / Departure



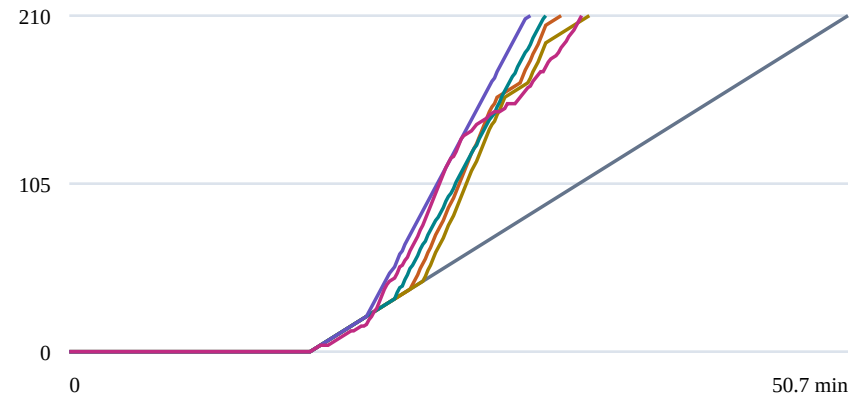
Vehicles waiting / Junction



Completed trips / Departure



Completed trips / Junction



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